

Ishan Roy

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Cornell MechE senior with hands-on experience in defense industry design engineering and aircraft rapid prototype development.

CORE COMPETENCIES: technical project leadership, design for manufacture, CAD, MATLAB modeling, rapid prototyping.

EDUCATION

Cornell University, David A. Duffield College of Engineering, Ithaca, NY

Master of Engineering in Aerospace Engineering.

Expected Graduation Dec 2027

Bachelor of Science in Mechanical Engineering.

Expected Graduation May 2027

Courses: Heat Transfer | Fluid Mechanics | System Dynamics | Mechanics of Engineering Materials | Dynamics of Flight Vehicles

GPA: 3.61/4.0.

EXPERIENCE

GE Aerospace, Defense Engines and Services. Lynn, MA

May – Sep 2026

Intern, Turbofan Turbojet (TFTJ) Mechanical Systems/Structures/Configs Design Engineering Team

- Perform structural hand-calculation analysis and prepare materials for technical design reviews, along with supporting process engineering efforts across mechanical systems, structures, and configuration teams.
- Implement a part classification framework adapted from GE's commercial business and present it at Tech Review, the first application of this approach within the defense organization, to support field failure mitigation and component life management.

Cornell Mechanical & Aerospace Engineering, Magnus Ultralight Project. Ithaca, NY

Jan 2025 – May 2026

Team Lead, advised by Professor Mason Peck and funded by industry partners.

- Led student team tasked with developing a Magnus Effect-powered ultralight electric aircraft for long-distance personal air travel.
- Designed a scale VTOL vehicle capable of consistent performance for data collection, utilizing SOLIDWORKS, ANSYS, and MATLAB.
- Fabricated and flight-tested prototype, conducting ten five-minute hovers while lifting 4 kg of payload (25% above initial goal).
- Designed, fabricated, and tested a dynamically balanced, gear-driven hub system for 1,500+ RPM Magnus Effect sphere actuation.

CU Design Build Fly (DBF) Project Team. Ithaca, NY

Design, build, test, and optimize a radio-controlled aircraft to compete in the AIAA Annual DBF Competition.

Senior Advisor and Analysis Subteam Co-Lead

June 2026 – present

- Provide technical oversight and lead onboarding. Establish and lead a new analysis subteam through workshops and mentoring.

Full Team Co-Lead

May 2025 – May 2026

- Led 40 members across four sub teams with a \$35k budget. Responsible for technical project leadership and daily operations.
- Instituted a rapid-prototyping process with tightened iteration cycles and elevated design review standards.
- Implemented improved composite vacuum-bagging process for the fuselage that raised vacuum pressure to 30 inHg to reduce excess epoxy and cut fuselage weight by 20% while maintaining strength and enabling sharper mold geometry than prior layouts.
- Spearheaded an overhaul of manufacturing and testing. 5 aircraft were designed built, and tested in a 22 flight test campaign (compared to 3 aircraft built and 3 test flights in prior years), resulting in the completion of multiple missions at competition (first time in 5 years), and a 33rd-place finish out of 200 schools (Cornell's best result in a decade).

Mechanical and Structural Subteam Member, Co-Lead

Sept 2023 – May 2025

- Led a 9-member team. Designed and fabricated DBF's first composite fuselage and landing gear, reducing aircraft weight by 45%.
- As member of subteam, made design improvements to nose cone and fabricated wooden fuselage.

Cornell Mechanical & Aerospace Engineering. Ithaca, NY

Jun 2026 – present

Teaching Assistant, Introduction to Aeronautics (MAE 3050), Professor Olivier DeJardins

- Create and grade homework and weekly quizzes, develop exams with course professor, and hold office hours.
- Develop and lead final design engineering project, where students will design and test a glider in a course wide competition.

Mohawk Mountain Alpine Ski Race Team. Cornwall, CT

Mar 2022 – present

Alpine Race Coach, U10-U16 athletes, Tri-State Alpine Ski Racing Association

PROJECTS

MATLAB Design Optimization Program

May 2025 – present

- Developed a MATLAB design optimization program to produce theoretical aircraft designs for AIAA DBF competition.
- Wrote the module to generate a high-fidelity structural model of the aircraft based on the design vector's input parameters.
- Tested algorithm and successfully generated designs scoring in the top 5% of several previous competition cycles.

TECHNICAL SKILLS

Software: SOLIDWORKS and MATLAB (advanced) | ANSYS (Fluent and Static Structural) (intermediate) | Excel (intermediate)

Programming Languages: Python (advanced) | HTML/CSS (intermediate). **Tools:** NumPy | pandas | scikit-learn | Aero Sandbox | Git.

Hardware: 3D printing and composite fabrication (advanced) | machining (mill, lathe, bandsaw) and soldering (intermediate)